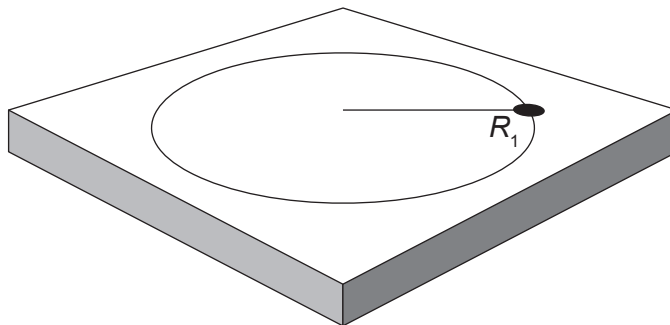


PAPER 1 / QUESTION 84

2 May 2023 / TZ2 (file label) / HL Paper 1 / original question 19 / 2223-6513

84. A mass on the end of a string is rotating on a frictionless table in circular motion of radius R_1 and undergoes an angular displacement of θ in time t .



The string tension is kept constant, but the angular displacement of the mass is increased to 2θ in time t . The radius of the motion changes to R_2 .

What is R_2 ?

- A. $\frac{R_1}{4}$
- B. $2R_1$
- C. $4R_1$
- D. $R_1 \times R_1$